

Interpretation of Event-Centered Pre/Post Graphs and Machine Learning Results

The event-centered time-series plots provide compelling visual evidence on how internal migration dynamics respond to different types of disasters in Turkey. By explicitly marking the disaster year (t_0), these graphs allow for a direct assessment of whether disaster events coincide with structural breaks in out-migration trends. Crucially, the visual patterns observed in the graphs align closely with the findings from the machine learning analysis, offering a coherent interpretation of migration responses across methods.

Major Earthquakes: Structural Breaks and Shock Effects

For provinces affected by major-disruptive earthquakes, such as Kahramanmaraş, Hatay, and Van, the graphs consistently reveal pronounced deviations from pre-event trends. In these cases, out-migration growth often drops sharply or becomes highly volatile following t_0 , indicating a clear structural break rather than a temporary fluctuation.

The machine learning results strongly reinforce this interpretation. While the Random Forest model outperforms the linear Ridge model overall, it still tends to underestimate the magnitude of the extreme negative migration responses observed after the 2023 earthquakes. This systematic underprediction suggests that major earthquakes do not merely amplify existing migration trends; instead, they fundamentally disrupt demographic momentum. The relationship becomes non-linear: rather than continuity, the disaster generates a localized shock in which typical migration patterns are constrained by large-scale housing destruction, economic collapse, and institutional disruption.

In this sense, major earthquakes act as threshold-crossing events, producing migration dynamics that cannot be accurately captured by linear trend extrapolation alone.

Minor Earthquakes: Persistence of Pre-Existing Trends

In contrast, provinces experiencing minor or non-disruptive earthquakes generally do not exhibit clear structural breaks in the event-centered graphs. Post-event out-migration growth typically remains within the range of pre-event variation and often reverts quickly to the existing baseline trend.

This visual stability is strongly supported by the machine learning findings. The Random Forest model achieves high predictive accuracy for these events, largely driven by the dominant importance of pre-event out-migration growth, which accounts for approximately 64 percent of total feature importance. This indicates that, in the absence of catastrophic destruction, historical migration momentum exerts strong inertia, rendering minor seismic shocks statistically insignificant in shaping medium-run migration outcomes.

Floods and Wildfires: Transient and Localized Effects

The event-centered graphs for flood and wildfire events display heterogeneous but largely temporary responses. While some provinces experience short-term spikes or drops in out-migration growth immediately following the event, these deviations are typically short-lived and do not translate into sustained trend changes.

The machine learning analysis corroborates this observation. Both flood and wildfire indicators receive very low feature importance (below one percent), suggesting that these events contribute minimally to explaining variation in post-event migration changes once baseline trends and province-specific characteristics are accounted for. These findings are consistent with recovery and return dynamics, which likely mitigate long-term displacement effects and distinguish floods and wildfires from the persistent structural damage caused by major earthquakes.

Overall Assessment: Linearity, Non-Linearity, and Threshold Effects

Taken together, the event-centered graphs and machine learning diagnostics demonstrate that disaster-induced migration in Turkey is highly non-linear and conditional on disaster severity. A clear threshold effect emerges. Below this threshold, including minor earthquakes, floods, and wildfires, migration dynamics are largely governed by historical trends and demographic inertia. Above this threshold, in the case of major-disruptive earthquakes, structural breaks occur and pre-event trends lose predictive power.

These findings provide strong justification for the event-based pre/post methodology employed in this study. They also highlight the limitations of treating disasters as uniform shocks within linear modeling frameworks. Instead, a segmented analytical approach—distinguishing between trend-continuing minor events and trend-breaking major events—is essential for accurately capturing the demographic consequences of disasters and for informing effective, targeted policy responses.